

Jothiswaran Arumugam

Forward Deployed Engineer | AI Engineer | Solutions Engineer

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Summary

4+ years building and deploying production AI/ML systems across healthcare, financial services, and startups. Customer-facing technical lead who owns problems end-to-end, from architecture scoping and prototyping through production deployment and live iteration. Shipped 25+ client engagements at an AWS Advanced Tier Partner; 13x AWS and 7x GCP certified.

Experience

Lead Data and AI Engineer

Jun 2025 -- Present

RevStar Consulting (AWS Advanced Tier Partner)

Tampa, FL (Remote)

- Automated a hospital data lake that replaced a quarterly USB-and-Excel reporting loop, parsing 11 SQLite tables to Parquet, deduplicating overlapping backups, and refreshing 6 QuickSight datasets through Step Functions in minutes.
- Shipped an 8-step agentic pipeline on Strands Agents SDK and Bedrock AgentCore that repairs and validates cardiac mesh files from STL through to quality-checked volumetric VTP, orchestrating PyMeshLab, SimVascular and ParaView as microservices on ECS Fargate.
- Cut a 962K-vector pgvector index from 7.5 GB to 3.1 GB with a quality-floored partial index, lifting topical nDCG@10 from 0.86 to 0.92 and restoring sub-second hybrid search after full-corpus queries had degraded to 59 seconds on the same instance.
- Engineered a Text2SQL analytics agent covering 50+ column schemas that converts natural language into Redshift SQL, built in Python on Claude Sonnet and Bedrock AgentCore with SQL injection prevention, PII guardrails, and multi-tenant isolation.
- Accelerated delivery timelines 60% across the practice by building internal RevStar QuickStart libraries (LLM-Ops, Bedrock-AgentCore, SageMaker-ML, Data-Lake); led code reviews and architecture design reviews, and mentored 8+ engineers on production AI workflows.

AI Engineer (Co-op)

May 2024 -- May 2025

IM Digital

Boca Raton, FL (Remote)

- Achieved sub-300ms retrieval latency for legal and product domains by building RAG pipelines with hybrid vector indexing on Qdrant, serving 1K+ daily queries via FastAPI on Cloud Run.
- Built model evaluation into CI/CD and lifted hallucination grounding scores 62% using LangSmith and RAGAS; fine-tuned LLaMA and Mistral via LoRA/QLoRA on Vertex AI, improving generation accuracy 37% and cutting token costs 35%.

GCP Cloud Engineer

Jul 2021 -- Aug 2023

Cognizant Technology Solutions

Hyderabad, India

- Reduced ML infrastructure setup time 80% across 15+ projects by automating provisioning with Terraform, GKE, and BigQuery; deployed Vertex AI pipelines through Jenkins CI/CD with Dataflow ETL to improve SLA adherence 60%.
- Mentored 30+ engineers on GCP provisioning, Vertex AI pipeline design, and Terraform module structure.

Projects

- Physician Candidate Matcher** (github.com/devjothish) - Physician-to-role matching that cut per-match inference from 180 LLM calls to 2 regardless of pool size, scoring 7 dimensions in pure Python and resolving specialty equivalence through NPI taxonomy on Postgres.
- AgentOps Lab** (github.com/devjothish/agentops-lab) - Framework-agnostic agent reliability toolkit: LangGraph agents emit OpenTelemetry trajectory traces, which agentops ingests, scores and visualizes so a change to an agent can be measured rather than argued about.
- Production Voice AI Agent** (github.com/devjothish/production-voice-ai-agent-livekit) - Real-time voice agent holding sub-800ms end-to-end latency at 50% lower cost than managed alternatives, with 3 independently swappable STT-LLM-TTS stages and LLM-as-judge evaluation on AWS CDK.

Technical Skills

Languages & Frameworks: Python, SQL, TypeScript, FastAPI, React, PostgreSQL, pgvector, PyTorch

LLM Systems in Production: Claude, GPT-4, Llama; RAG, hybrid retrieval, embeddings, vector search; AI agents, multi-agent orchestration, tool use, Model Context Protocol (MCP); LangChain, LangGraph, CrewAI, Strands Agents SDK; prompt engineering, guardrails, PII redaction, human-in-the-loop review; LLM evaluation (RAGAS, LangSmith, LLM-as-judge), QLoRA/LoRA fine-tuning

AWS & Data Platform: Bedrock, Bedrock AgentCore, SageMaker, Lambda, ECS Fargate, Step Functions, S3, API Gateway, DynamoDB, OpenSearch, Redshift, Glue, Athena, Lake Formation, QuickSight; Terraform, AWS CDK, Docker, CI/CD, OpenTelemetry; GCP (Vertex AI, BigQuery), Azure (OpenAI Service)

Certifications

AWS (13x - Golden Jacket): GenAI Developer Professional, Solutions Architect Professional, DevOps Engineer Professional, Advanced Networking Specialty, Machine Learning Specialty, Security Specialty, Developer Associate, Machine Learning Associate, Data Engineer Associate, Solutions Architect Associate, CloudOps Engineer Associate, AI Practitioner, Cloud Practitioner

Google Cloud (7x): Professional Machine Learning Engineer, Professional Cloud Architect, Professional Data Engineer, Professional Cloud Developer, Professional DevOps Engineer, Professional Database Engineer, Associate Cloud Engineer

Education

Florida State University

Aug 2023 -- May 2025

Master of Science in Computer Science, GPA: 3.72/4.0

Tallahassee, FL

Rajalakshmi Engineering College

Aug 2017 -- May 2021

Bachelor of Engineering in Computer Science, GPA: 3.5/4.0

Chennai, India